



The Forgotten Crystal Radio – Revisited

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You may be wondering about this article on a crystal radio in an era of digital broadcasting and digital audio. Crystal radios were quite popular and played an important role in communication from 1920s to 1940s. The number of people using crystal radio substantially reduced after transistor radios came into the market in 1960s. Even though the crystal radio is a hundred years old, it is still living as a hobby in several countries where medium wave (MW) and short wave (SW) amplitude modulated (AM) transmission are available on air.

Several crystal radio forums and groups are still active and their highly advanced crystal radios can receive radio stations several hundred kilometres away. Search on Wikipedia, You Tube, and Google provides a lot of information on crystal radios and it is a good place to learn more about it.

Crystal radio is the pioneer of widely used radio receivers, though it is almost forgotten now in most countries including India. Construction of crystal radio was a thrilling and fascinating hobby of all radio hobbyists of those days (including two of the authors who were teenagers in 1960s). A crystal radio needs no battery or power and works on the microvolts of radio signals received from its antenna.

India has more than 300 medium wave radio stations located in major cities and towns. The presence of these radio stations offers an opportunity to construct crystal radios for electronics hobbyists, including school and college students. Our neighbouring countries like Sri Lanka, Malaysia, and Singapore have already shut down all amplitude modulated (AM) medium wave stations and switched to frequency modulated (FM) transmission. So, the present-

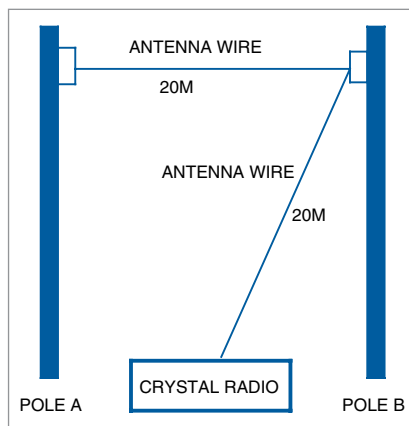


Fig. 1: Outdoor antenna

day electronics students, teachers, and hobbyists in India are really lucky as they can still enjoy amazing power of the tiny crystal radio since plenty of AM stations are available to tune in to.

But how many present-day electronics hobbyists, school students, and engineering students are even aware of this excellent piece of electronics is still a question. Hence, this article is intended to reintroduce this simple radio and to revive interest in crystal radio among school students, and several thousands of engineering students passing out from thousands of engineering colleges/Universities across India. The circuits provided here are not our own but a rediscovery of the knowledge present during the 1930s.

A crystal radio needs no battery or power and works on the microvolts of radio signals received from its antenna. It works non-stop (24 × 7) as long as the transmission is on air. The construction of crystal radio is remarkably simple with just four components required to make it. It needs an antenna, a good earth connection, a germanium diode, and a high-impedance headphone.

You will be fascinated by the working of crystal radio provided you live

in or near a city where a MW wave radio station is in operation. However, those who live far away from a MW station can also receive the radio signal with the help of a tall and long aerial (antenna) and good earth connection. Although long wave (LW) and short wave (SW) crystal radios were also constructed by several people, this article focuses on crystal radio that receives MW amplitude modulated transmission as we have a lot of MW stations in India to receive.

Before constructing the crystal radio, please understand that this crystal radio does not have an amplification circuit. So, the sound output totally relies on how effective the antenna and earth are, besides the power of your local radio station and the distance between your radio and the transmitting tower. A minimum of 18-metre single-core or multistrand hookup copper wire (14/36), or 20SWG enameled copper wire antenna, erected in open terrace will work well.

It is better to know about the transmitting power of the nearby radio station. In India, some cities like Dibrugarh (Assam), Jammu, Srinagar, Nagpur, Imphal, Jalandhar, Lucknow have 300kW high-power transmitters and other cities like Ahmedabad, Alleppey, Ajmer, Bengaluru, Mumbai, Kolkata, Chennai, Delhi, Dharwad, Hyderabad, Indore, Jabalpur, Siliguri have 200kW transmitters. Those who live in and around these cities will get good reception from local stations with the crystal radio.

Further, there are 100kW, 50 kW, and 20kW MW transmitters in operation in various cities and towns. Those who live near low-power stations (20kW) can use an amplifier as described in this article. Visit www.allindiaradio.gov.in for the list of all radio stations with their power in kilowatts. Those who live near